

Module 1 Practice: Exploratory Data Analysis Part-1

Setup: For this practice, we will use the penguins dataset from the Seaborn library. Open a new Jupyter Notebook and run the following code to get started:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Load the dataset
df = sns.load_dataset('penguins')
```

Task 1:

Using the pandas functions covered in class, answer the following:

- What is the total number of samples (rows) and potential features (columns) in this dataset?
- Display the first 7 rows of the dataset to understand the structure.
- Display a random sample of 5 rows.

Why might it be better to look at a random sample rather than just the top rows? Answer in two lines

Task 2:

Before analyzing the data, we must check for missing information.

- Write the code to calculate the total number of missing values for each column.
- Which feature contains the highest number of missing values?

Task 3:

Duplicate entries can hamper model performance.

- Check if there are any duplicate rows in the penguins dataset. How many duplicates are there?
- Write the code to permanently remove any duplicate rows from your DataFrame.

Task 4:

Let's profile the dataset to understand its distributions.

- Use the appropriate method to print a concise summary of the DataFrame, including the data types of each column and non-null counts.
- Generate a statistical summary for the numerical columns.
- Based on your statistical summary, what is the maximum **body_mass_g** of a penguin in this dataset? What is the average (mean) **flipper_length_mm**?

Task 5:

Create a visualization to understand the distribution of penguin species.

- Use Seaborn to create a count plot for the species column.
- Which species is the most common in this dataset?

Task 6:

- Write the code to calculate the percentage distribution of the island column

Task 7:

Let's visualize the gender distribution among the surveyed penguins.

- Extract the value counts for the sex column.
- Create a pie chart using Matplotlib showing the male vs. female distribution.
- Ensure the pie chart includes percentage labels with one decimal point (example 50.5%).

Add a title to your chart.